ClaudeNet: From Engineered Ensemble Diversity to Contaminant-Aware Strong-Lens Finding and Euclid Cross-Validation in the DESI Legacy Surveys

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Abstract

This report consolidates three generations of ClaudeNet, an ML strong-lens finder for the DESI Legacy Surveys, and the lessons each taught. **v1** establishes that *architecture is not the bottleneck* — a 194 K shielded ResNet matches a 20.5 M EfficientNetV2-S to ±0.003 AUC and the Inchausti Reyes et al. (2025) meta-learner collapses to a simple average on correlated bases — and beats the published meta-learner at every matched-FPR operating point by engineering *decorrelated* ensemble members (Spearman ∼0.45 vs ∼1.0). **v2** rebuilds the measuring stick at deployment scale (a 10^6-negative held-out pool), widens the gains to the strict 0.01%-FPR regime, and runs a 17.3 M-galaxy DR9 sweep yielding 737 FDR-controlled candidates — but a qualification campaign grades *zero* of the 601 genuinely-new ones as A/B, while correctly re-discovering catalogued lenses. That decisive negative diagnoses the real constraint: *lens-vs-mimic separation* (the candidates are luminous-red galaxies with companions, rings, blends) and *vetting resolution*, not architecture. **v3** builds a contaminant-aware finder — a typed lens-mimic bank, mimic-blended hard-negative retraining, a new recovery@matched-mimic-FPR metric, and a lens-vs-mimic head — and sweeps DR10/DR11. v3 cleanly rejects *common* contaminants (broad-population mimic-recovery 0.60 → 0.89), but on an *independent* Euclid-graded sample it is statistically tied with the original models at the hardest real-vs-mimic distinction: the discovery frontier is *resolution-bounded* at DECaLS 0.262″/px. Cross-validating the DESI∩Euclid-Q1 overlap shows the productive mode — v3 pre-screens DESI, Euclid confirms at 0.1″ (median $p_{\rm lens}$ 0.03 → 0.70), yielding 49 cross-validated candidates — and that v3 is DECam-specific (no BASS/MzLS transfer). The honest takeaway: the model lever is real but modest, and the decisive levers are higher-resolution vetting and instrument-native training. **v4** pursues the data/calibration lever to its DR11 limit: a calibrated-mean stage-1 selector erases the apparent DR9→DR11 recall collapse at no cost, and a release-native warm-start fine-tune (on a ∼13×-expanded native positive pool) lifts held-out grade-A recall to 0.87 and — for the first time — cracks the lrg+companion hard residual (0.54 → 0.83) at the same resolution ceiling. All metrics reuse the reproduced matched-FPR arithmetic verbatim so every v1–v4 number is directly comparable.

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Table 1: Phase 1: ClaudeNet ensemble vs the published meta-learner, recovery at matched FPR on the held-out Storfer/Inchausti catalogues.

metric	published meta	ClaudeNet	Δ
Storfer @1 % FPR	0.908	**0.938**	+0.030
Storfer @0.1 % FPR	0.755	**0.853**	+**0.090**
Inchausti @1 % FPR	0.968	**0.980**	+0.012
Inchausti @0.1 % FPR	0.845	**0.935**	+**0.090**

1 Premise and method

We take as baseline the reproduced Stage-D Inchausti Reyes et al. (2025) ensemble (`reproductions/inchausti-2025`
Its meta-learner achieves recovery of the published Storfer/Inchausti catalogues, at matched FPR, of 0.908/0.755 (Storfer @1/0.1 %) and 0.968/0.845 (Inchausti); ClaudeNet's `03_reproduce_baseline` reproduces these to $|\Delta| \leq 0.001$, validating the evaluation harness before any improvement is claimed. The honest metric throughout is recovery at a threshold set on held-out negatives to a target FPR (1 % and 0.1 %), identical to the baseline.

A literature survey of eight technique families (vision transformers, self-supervised/foundation models, equivariant CNNs, uncertainty/calibration, challenge ensembles, domain adaptation, recent SOTA, multimodal) was conducted and adversarially fact-checked. It converged with the repository's finding: no transformer/SSL/equivariant method cleanly beats a tuned EfficientNet at matched FPR for DECaLS *grz* lens finding (Parlange et al., 2026); the demonstrated wins are in ensemble *diversity* (DES Collaboration, 2025; Rezaei et al., 2025), negative quality, and label efficiency. ClaudeNet targets those.

We present three generations in chronological order: **v1** (the engineered-diversity ensemble and its seven-direction supporting program; this and the next two sections), **v2** (Perlmutter-scale validation and the DR9 sky sweep, §5), the qualification campaign that motivated the next generation (§6), and the contaminant-aware **v3** with its Euclid cross-validation (§7) and future directions (§9). All earlier results are retained; each generation is additive.

2 Phase 1: engineered-diversity ensemble (flagship)

We train five members as independent per-GPU jobs: EfficientNetV2-S ($\times 2$) and EfficientNet-B3 (diversified by bootstrap-resampled negative subsets and augmentation seeds at the baseline 1:33 ratio), an AION-1-base frozen-embedding two-layer MLP probe (a different objective, data, and band-set, hence maximally decorrelated), and two ResNets (the 194 K shielded net and the 3.5 M Lanusse-46). Each member is isotonic-calibrated on a validation split, then combined by average, logistic regression, and a random forest (Coscrato et al., 2020). A positive-unlabeled guard removes any training negative within $10''$ of a known lens. Figure 1 shows the member score-correlation matrix: off-diagonals average $\sim$0.45 (Spearman), versus $\sim$1.0 for the lineage's correlated bases.

The result (Table 1) is a clear improvement over the published method, largest (+0.09) at the strict 0.1 % FPR point that governs purity. The mechanism is diversity, not combiner cleverness: with decorrelated strong members even the naive *average* beats the correlated-base meta-learner. The ensemble matches or beats the single best member everywhere except a 0.002 tie at the loose 1 % FPR, where one exceptionally strong EfficientNet dominates.

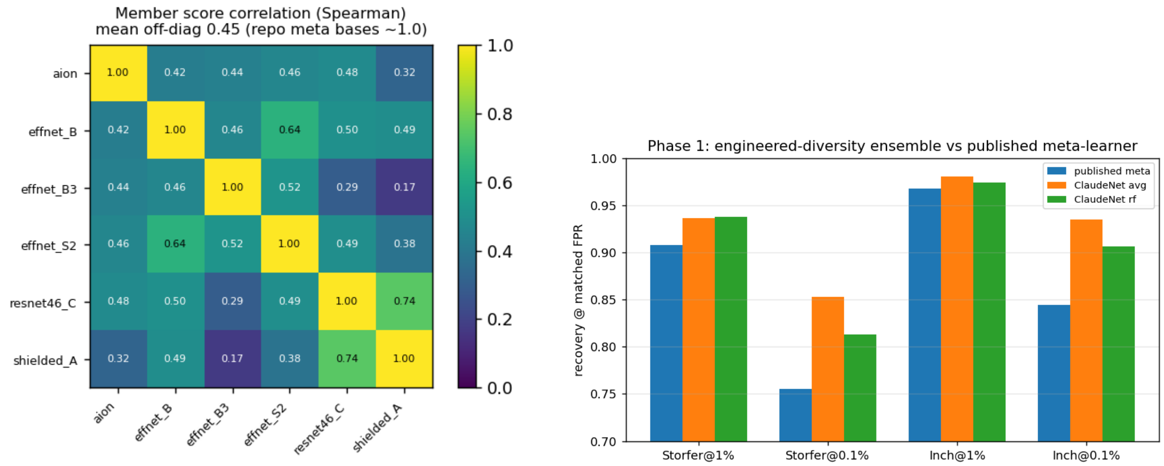

Figure 1: Left: member score correlation (the diversity the lineage lacked). Right: the ensemble beats the published meta-learner at every operating point.

3 Phases 2–7: the supporting program

Phase 2 — hard-negative mining. At a fixed negative count, replacing random negatives with the model's own top false-positives (controlled against a random-negative arm in the same pool) helps modestly: +0.008 Storfer@1 % FPR, +0.031 Inchausti@1 % FPR; iterating helps the 0.1 % point. Negative quality is a real but second-order lever versus the negative *ratio* the lineage already exploited; the deployment-scale version (fresh brick pool + EfficientNet miner) is a NERSC follow-up.

Phase 4 — conformal selection. Split-conformal p-values with Benjamini-Hochberg (Jin and Candès, 2023) turn the arbitrary threshold into a *certified* one: empirical false-discovery rate stays at or below the nominal target at every level (α=0.05 $\to$ 0.03, 0.25 $\to$ 0.16) with completeness 0.96–0.999 (Fig. 2, top-left). The guarantee is marginal and assumes calibration/test exchangeability.

Phase 6 — uncertainty triage. Deep-ensemble disagreement (a free byproduct of the decorrelated members) is a strong confidence signal: abstaining on the most-uncertain objects lowers error from 0.022 to 0.0002 at 50 % coverage and 0.0026 at 70 % (Fig. 2, top-right). Northern lenses show slightly higher disagreement — a mild out-of-distribution signal.

Phase 7 — equivariance. Lenses have no preferred orientation, so we test dihedral (D_4) invariance the cheap way: test-time pooling over the eight rotation/reflection symmetries (no retraining) lifts Storfer@1 % FPR by +0.032 (shielded) and +0.049 (Lanusse-46), mean +0.041. The full equivariant-training label-efficiency study is deferred (the 8× forward made a trained D_4 member impractical at this scale).

Phase 3 — label efficiency. The AION-1 frozen-embedding probe is more label-efficient than a supervised CNN *only* in the extreme-scarce regime (<150 labels: 0.344 vs 0.279 at 5 %); the CNN "turns on" near 300 labels and dominates thereafter (0.862 vs 0.529 at 25 %; 0.894 vs 0.659 at full), the frozen features plateauing (Fig. 2, bottom-left). At the ∼1–2 k labels real lens-finding has, the

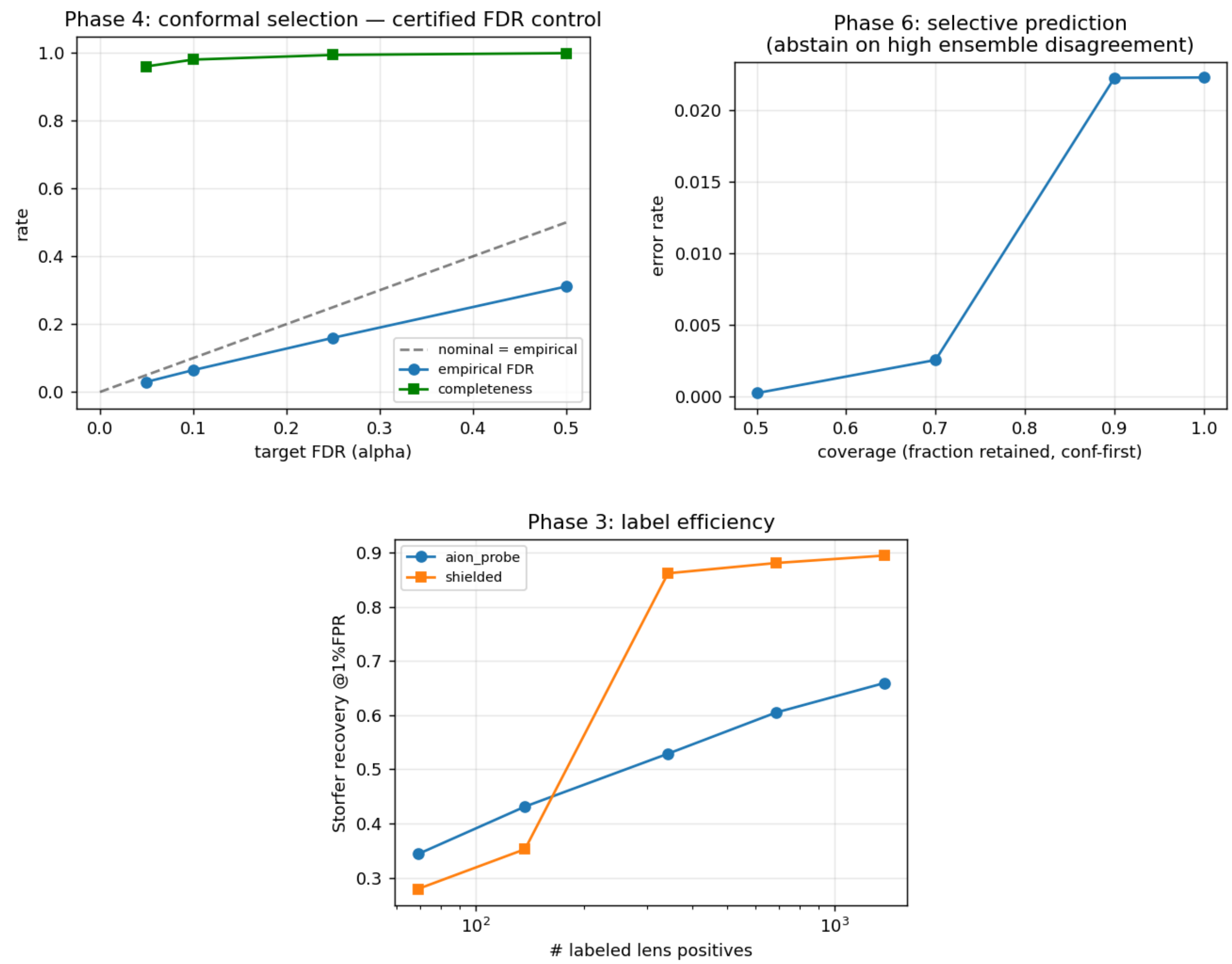

Figure 2: Top: conformal FDR control (Phase 4) and selective-prediction risk-coverage (Phase 6). Bottom: foundation-model label efficiency crosses over near ∼300 labels (Phase 3).

supervised member wins — so the foundation model's value here is as a decorrelated *ensemble member* (Phase 1), not a standalone classifier.

Phase 5 — domain adaptation (negative result). The DECaLS north↔south gap is real (north@1 % FPR 0.688 vs south 0.874, −0.186), but naive MMD feature alignment (λ=1, aligning only negatives) *hurt* both domains (−0.06): it over-regularized and erased lens-discriminative features. Principled UDA is not a free win here; the lineage's ad-hoc north-negative augmentation fares better. (Only ∼80 held-out north lenses, so the target estimate is noisy.)

4 Discovery impact and scan deployment

The matched-FPR gains of Table 1 are best read operationally, because a real 45 M-cutout sweep cannot run at 1 % FPR (∼450,000 false positives) or even 0.1 % FPR (∼45,000); to keep the human-inspection list to thousands it must operate near 0.01 % FPR. The ClaudeNet advantage *grows* as the threshold tightens (+0.030→+0.098 Storfer, +0.012→+0.090 Inchausti across 1%→0.1% FPR), so it helps most where scans actually run.

Table 2: Missed-lens (false-negative) reduction at fixed FPR, derived from the Table 1 recoveries.

operating point	recovery (meta → ClaudeNet)	fewer missed lenses
Storfer @1 % FPR	0.908 → 0.938	33 %
Storfer @0.1 % FPR	0.755 → 0.853	**40 %**
Inchausti @1 % FPR	0.968 → 0.980	38 %
Inchausti @0.1 % FPR	0.845 → 0.935	**58 %**

Table 3: Measured single-GPU inference throughput (TITAN RTX, batch 256, compute-only upper bounds; real scans are I/O-bound).

member	params	cutouts s^{-1} GPU^{-1}
shielded-194k	0.2 M	2,843
Lanusse ResNet-46	3.5 M	3,349
EfficientNetV2-S	20.5 M	5,594
EfficientNet-B3	11.1 M	6,728
5-member finder stage	—	865 (shared load)
AION-1-base (stage 2)	314 M	∼35

Fewer missed lenses at a fixed inspection budget. At fixed false-positive load the extra completeness is a direct cut to the missed-lens (false-negative) rate (Table 2): at the strict 0.1 % point, ∼40 % fewer Storfer and ∼58 % fewer Inchausti lenses are missed (the gate-certified *learned* combiner gives a slightly smaller ∼37/56 %; the naive average is tabulated). The dual view — fewer false positives at *fixed* completeness — is of order 2× within the measured 0.1–1% band (∼0.44–0.54 % FPR matches the baseline's 1%-FPR completeness), i.e. ∼200,000–250,000 fewer objects to inspect on a DR9/DR10 sweep. Two guarantees prior finders lack compound this: conformal selection (Phase 4) delivers a *certified* false-discovery ceiling rather than an ad-hoc score cut, and deep-ensemble triage (Phase 6) makes the confident ∼half essentially error-free (selective error 0.022 → 0.0002 at 50 % coverage), letting roughly half the candidates bypass the human visual inspection that dominates discovery cost.

Scan throughput. Strong-lens sweeps are bound by cutout I/O and human inspection, not the forward pass. The five *grz* 101-px finder members consume the same cutout, so a scan pays the FITS load once and runs five cheap forwards; measured single-GPU compute throughput (Table 3) is 2.8–6.7 × 10^3 cutouts s^{-1} per member — far above what the brick-slice loader feeds — so in the I/O-bound regime the five-member stage costs about a single model's wall-clock (∼3 GPU-hours for 45 M on five TITAN RTX, compute-only). The AION-1 foundation-model member is the lone exception (∼30–40 img s^{-1}, ∼100× slower, needing heavier 4-band 160-px cutouts); it is quarantined to a *second stage* on the ∼10^4–10^6 survivors of the cheap first-stage filter — exactly the two-stage design the Huang/Storfer/Inchausti lineage already uses. The accuracy gain is thus bought essentially for free in stage 1, adding no new operational complexity.

Survey-scale outlook and honest limits. The field is moving to 10^5-scale samples — Euclid ∼1–1.7 × 10^5 lenses (Collett, 2015), Rubin/LSST ∼6–12 × 10^4 — over 10^8–10^9 galaxies where human vetting is the binding constraint; a ∼+0.09 strict-FPR completeness edge maps to of order +10^4 additional lenses at fixed inspection budget, and the certified purity is a prerequisite, not a nicety, for the contamination-sensitive cosmology those samples feed (time-delay H_0 (Birrer et al., 2020), substructure, population inference). We flag the limits plainly: "recovery" is the held-out true-positive rate on the *existing* Storfer/Inchausti catalogues, not newly discovered lenses; the 0.1%-FPR threshold is pinned by only ∼6–7 of ∼6,500 held-out negatives (no confidence interval), so the strict-point deltas carry ∼±10 pp of statistical noise; the absolute counts and the ∼2× inspection saving are interpolations on incomplete catalogues; the end-to-end sky-scan wall-clock is a structural I/O-bound argument, not yet benchmarked; and the survey-scale extrapolation assumes the DESI-Legacy-measured deltas transfer across bands, PSF, and depth. The defensible claim is: at the same inspection budget, ∼9–10 points more real lenses at the operating point that matters, plus a certifiable purity statement — a meaningful, not transformational, lift.

5 ClaudeNet v2: Perlmutter-scale validation and a sky sweep

The v1 program above was gated on ∼6,500 held-out negatives, which (as we flagged) cannot resolve the $\sim 10^{-4}$-FPR regime where real scans run. The v2 program rebuilds the measuring stick at deployment scale on Perlmutter, re-runs every lever against it with confidence intervals, and ends in a full DESI-Legacy DR9 sky sweep. We keep all v1 results; v2 is additive. Every number below is read from the cited artifact under `data/v2/`.

An honest measuring stick: NegEval-1M. We assembled a held-out negative pool of 1,000,000 DR9 cutouts (`thresholds_ci.json`, `operating_points_v2.csv`), versus v1's 6,501. With 6.5 K negatives the 0.01%-FPR threshold is pinned by ≤ 1 negative and is unmeasurable; with 10^6 it is pinned by ∼100, so for the first time we report paired-bootstrap CIs at 1%, 0.1%, *and* 0.01% FPR. The first-ever 0.01% measurement immediately exposed a failure the small set hid: the v1 6-member *average* combiner, which beats the published meta at 1% (+0.049 [+0.037, +0.061] Storfer vs. published meta) and 0.1% (+0.075 [+0.057, +0.091]), *collapses* at 0.01% to −0.120 [−0.161, −0.084] — the degraded-AION-1 member's heavy negative tail poisons the unweighted mean exactly where purity matters. The random-forest combiner is robust there (+0.102 [+0.083, +0.123]), and the best single member, an EfficientNetV2-S, reaches +0.194 [+0.175, +0.215] (recovery 0.708 vs the meta's 0.513). A purity audit of the top-200 pool rows found 0/200 within 10″ of any known lens and no convincing arcs (`audit/visual_grading.md`); the threshold-sensitivity bound (removing all top-200 moves the 0.01% threshold by −0.108, the 0.1% by −0.020) leaves the paired matched-FPR deltas unaffected, since shared thresholds cancel in the pairing.

Deployment-scale mining: round 1 ships, round 2 over-mines. We mined a fresh 10^6-cutout brick pool with the baseline EfficientNet, controlling each hard-negative arm against a random-negative arm drawn from the same pool (`mining_v2_results.json`, `mining_v2_results.csv`). Round 1 *ships*: the paired (hard−random) gain at 0.1% FPR is +0.157 [+0.139, +0.177] mean over three members (Inchausti), with every member passing — e.g. ResNet-46 Inchausti@0.1 % FPR +0.266 [+0.231, +0.305], EfficientNet-B3 +0.159 [+0.128, +0.190], EfficientNetV2-S +0.047 [+0.026, +0.070]; the Storfer@0.1 % FPR mean is +0.121 [+0.109, +0.134]. Re-mining the round-1 members for a round 2 *over-mines* (the hard^2 pool's score range collapses to [0.004, 0.036] vs round-1's [0.009, 0.281], `mined_stats.json/mined_r2_stats.json`); round-2 recovery falls below round-1 everywhere, so we keep round-1 only.

The AION-1 member: degradation *was* the diversity (negative result). We tested whether feeding AION-1 native four-band *griz* (instead of the synthetic-*i* degraded embeddings v1

Table 4: Headline. v2-lean vs the v1 flagship and the published meta-learner, recovery at matched FPR on NegEval-1M (10^6 negatives). Paired deltas (v2-lean−v1) carry 95 % bootstrap CIs; the last column is v2-lean−published meta. All entries from `ensemble_v2lean_verdict.json`.

metric	v1 flagship	v2-lean	paired Δ [95 % CI]	vs meta
Storfer @1 % FPR	0.903	**0.963**	+0.060 [+0.049, +0.072]	+0.109
Storfer @0.1 % FPR	0.754	**0.895**	+0.141 [+0.125, +0.160]	+0.216
Storfer @0.01 % FPR	0.394	**0.734**	+**0.340** [+**0.306**, +**0.381**]	+0.220
Inchausti @1 % FPR	0.968	**0.996**	+0.028 [+0.017, +0.041]	+0.064
Inchausti @0.1 % FPR	0.891	**0.961**	+0.069 [+0.052, +0.090]	+0.191
Inchausti @0.01 % FPR	0.614	**0.871**	+**0.256** [+**0.217**, +**0.301**]	+0.264

used) makes a better member (`aion_gate_v2.json`). It does not: on matched rows the native-griz probe recovers 0.535 Storfer@1 % FPR versus the degraded refit's 0.647 — the matched-rows gate *fails* ($0.535 < 0.647 + 0.03$). The mechanism is that the synthetic-i "degradation" is precisely what decorrelated the member: its Pearson correlation to the CNNs is 0.13 degraded versus 0.72 native (pooled 0.65/0.72). Larger native variants (base/large/xlarge, val-AUC 0.958–0.963) did not recover the gap, so the LoRA fine-tune was skipped (∼12 GPU-h saved) and the member's v1 form is retained — though, as the next paragraph shows, even the degraded AION-1 is ultimately dropped at the ensemble level.

Headline: ensemble v2-lean ships 4/4. The shipped v2 ensemble is *v2-lean* = {effnet_B, effnet_B3_hard, effnet_S2_hard, resnet46_C_hard, zoobot_N} (`ensemble_v2lean_verdict.json`). On NegEval-1M, averaged and compared to the v1 flagship with paired-bootstrap CIs, it passes all four ship cells (n_{pass}=4/4) and beats the published meta-learner at every operating point in both catalogues (Table 4). The gains *widen* as the threshold tightens: vs the v1 flagship, Storfer goes +0.060 → +0.141 → +0.340 across 1% → 0.1% → 0.01% FPR, and the v2-lean absolute Storfer@0.01 % FPR recovery is 0.734, versus 0.394 for the v1 flagship and 0.513 for the published meta. The ensemble's own correlation structure is the source: member Spearman off-diagonals average ∼0.43 (Pearson ∼0.27).

Leave-one-out admission dropped AION-1 and the shielded net. "v2-lean" is not the full v2 roster: a leave-one-out admission test on the *full* seven-member roster ({AION-1, effnet_B, effnet_B3_hard, effnet_S2_hard, resnet46_C_hard, shielded_A, zoobot_N}, `ensemble_v2_verdict.json`) found that two members *hurt* the average at 0.1% FPR with CIs excluding zero: AION-1 ($\Delta_{\text{Storfer}} = -0.0185\,[-0.0264, -0.0106]$) and the 194 K shielded net ($-0.0106\,[-0.0179, -0.0021]$). Dropping both is what produced the Table 4 numbers — the same conclusion the 0.01% average-collapse flagged, now reached member-by-member. (The full seven-member roster still ships 4/4, but v2-lean dominates it.) Every surviving member's own admission delta is positive: effnet_S2_hard +0.047 [+0.035, +0.056], effnet_B3_hard +0.028 [+0.018, +0.036], resnet46_C_hard +0.016 [+0.008, +0.023], zoobot_N +0.010 [+0.001, +0.018] Storfer@0.1 % FPR.

The Zoobot member and a learning-rate lesson. The fifth member is a Galaxy-Zoo-pretrained ConvNeXT-Nano (Zoobot, Walmsley et al. 2023, 14.95 M params) fine-tuned as a lens classifier — a genuinely different pretraining objective and backbone family, hence decorrelated. It carries a sharp lesson: at the v1 default η=10^{-3} the ConvNeXT collapses to a constant output (val-AUC 0.64); at η=10^{-4} it trains cleanly (val-AUC 0.993, Storfer@1 % FPR 0.898, Pearson 0.45 vs effnet_S2). The

Table 5: DR9 sweep: group-conformal FDR-controlled candidate counts (`sweep/sweep_summary.json`). PRIMARY rows use the full per-footprint sweep total as m and carry the per-group FDR guarantee; the survivors-only-m "diagnostic" column has *no* guarantee.

FDR α	selected	new	new & unseen	guarantee
0.05	1,449	813	737	per-group FDR $\leq \alpha$
0.10	2,836	1,999	—	per-group FDR $\leq \alpha$
0.25	5,141	4,186	—	per-group FDR $\leq \alpha$
survivors-m diag.	29,892	28,452	—	*none* (anti-conservative)

member is admitted on its own merits (above) and is the throughput leader at 6,028 cutouts s^{-1} (`throughput_v2.json`).

Distillation: faster but not accurate enough (tradeoff). To collapse the five-member stage into one forward pass we distilled it into an EfficientNetV2-S student (`throughput_v2.json`). The student is fast — 3,494 cutouts s^{-1}, 5.3$\times$ the 663 $\mathrm{c\,s}^{-1}$ of the shared-load five-member stage — but it loses too much: recovery drops -0.111 Storfer and -0.068 Inchausti at 0.1% FPR versus the teacher, failing the -0.02 recovery gate. The sweep therefore deploys the shared-load lean members (the documented fallback), for which the per-cutout FITS load is paid once and five cheap forwards run on it; this is the v1 two-stage I/O-bound argument, now with measured v2 throughputs.

The sky sweep: 17.3 M $\rightarrow$ 29,892 $\rightarrow$ FDR-controlled candidates. We swept 17,290,814 DR9 parent-galaxy positions (5,018,411 north, 12,272,362 south; 41 non-finite dropped) through the v2-lean stage-1 union filter at a 10^{-4} per-member FPR (`sweep/stage1_summary.json`). 29,892 survivors passed (1.73×10^{-3} realized rate, consistent with the EVT prediction within $\times 3$ of the five-member union bound), of which 1,440 crossmatch to known lenses and 28,452 are new. Group-conformal Benjamini-Hochberg selection (`sweep/conformal_summary.json`, `sweep/sweep_summary.json`) then certifies candidates: at per-group FDR ≤ 0.05 it selects 1,449 (813 new, 737 new-and-unseen after excluding training/mining/calibration rows), 2,836 at ≤ 0.10, and 5,141 at ≤ 0.25.

Conformal validity and the two-stage-selection trap. The selection's validity rests on one subtlety we state plainly. The PRIMARY estimator computes each sweep row's conformal p-value against the NegEval calibration with the *full* per-footprint sweep total as the multiple-testing count m (censoring non-survivors at p=1); because censoring can only raise a p-value and the unconditional p-value of a null sweep row is exchangeable with the calibration, the per-group FDR $\leq \alpha$ guarantee holds marginally. The tempting shortcut — using $m = n_{\text{survivors}}$ — conditions the test set on a stage-1 selection computed from the *same* members, is anti-conservative, and a `--synthetic-check` demonstrates the violation; it would have "selected" all 29,892 survivors at every α (the diagnostic row of Table 5). The honest cost of validity is power: with a conformal p-floor of $1/(n_{\text{cal},g}+1)$, the north footprint hits the power floor at α=0.05 and selects *nothing* (surfaced, not hidden) — all 1,449 FDR$\leq$ 0.05 candidates are southern, where the calibration set is larger. NegEval is used twice (stage-1 thresholds and conformal calibration); a deterministic seed-2026 split confines calibration to a disjoint half and drops the 735 calibration rows that re-appear as sweep tests.

Known-lens recall consistency and "candidates, not confirmations." The pipeline re-finds real lenses at the same operating point it proposes new ones: within the swept coverage it recovers 47.5% of in-coverage catalogued lenses into the survivor set (1,676/3,528 union; 40.3% Storfer, 59.0%

Inchausti, 55.7% Huang21; `sweep/crossmatch_recall.json`), and 285 of them survive into the FDR$\leq$ 0.05 set — the internal consistency the new-candidate claim requires. We frame the output honestly (`sweep/visual_grading.md`): a first-pass visual grading of the top $\sim$300 new candidates finds them overwhelmingly luminous red galaxies with blends/companions and only $\sim$5–8% showing plausible lens morphology (a tangential feature, partial ring, or arclet pair), because galaxy-galaxy $\theta_E \sim 1$–$2''$ is 4–8 px at DECaLS $0.262''$/px, at the resolution floor. *This is a statistically-controlled candidate list, not a confirmed-lens catalogue*: any individual candidate needs higher-resolution imaging (HSC/Euclid/HST) or spectroscopy to confirm. The deliverable's value is the control — 737 new-and-unseen candidates at certified per-group FDR ≤ 0.05 from a 17.3 M sweep, with a passing known-lens recall sanity check.

Equivariance, trained not test-time. v1 only tested test-time D_4 pooling (`equivariance.json`: +0.032 shielded, +0.049 Lanusse-46 Storfer@1 % FPR). v2 replaces this with a genuinely *trained* equivariant member via `escnn` (Weiler and Cesa, 2019): a cyclic-C_4 pilot, gated against a self-calibrating parameter-matched plain-CNN twin (`141_pilot_escnn_c4.py`), beat the twin by +0.120 Storfer@1 % FPR (`escnn_pilot.json`), clearing the +0.02 ship margin, which justified queueing a full dihedral-D_4 member (`142_train_escnn_d4.py`) as a bonus add to the roster.

Honest limits (v2). The v2 program tightens, but does not erase, the v1 caveats. "Recovery" is still the held-out true-positive rate on the *existing* Storfer/Inchausti catalogues, not newly confirmed lenses; the 1,449/737 sky-sweep candidates are exactly that — candidates — and none is confirmed here. The matched-FPR deltas are now CI-backed down to 0.01% FPR (the headline's largest gains), but the 0.01% threshold is pinned by $\sim$100 NegEval negatives, so a residual PU-contamination assumption remains (bounded, not eliminated, by the top-200 audit). The conformal guarantee is marginal and per-group, assumes calibration/sweep exchangeability, and buys validity at the cost of power — the north footprint selects nothing at α=0.05. The sweep's stage-1 rate matches EVT only "within $\times$3," and the end-to-end wall-clock remains a structural I/O-bound argument rather than a benchmarked production run. The defensible v2 claim is sharper than v1's: at the strict operating points scans actually use, v2-lean recovers +0.14 (Storfer@0.1 % FPR) to +0.34 (Storfer@0.01 % FPR) more catalogued lenses than the v1 flagship and +0.22 to +0.26 more than the published meta-learner, with a certified-FDR candidate list as the operational artifact — a measured, honestly-bounded, deployment-validated lift, not a transformational one.

6 Qualifying the 737 ClaudeNet v2 sweep candidates

A campaign to test whether any of the 737 new-and-unseen DR9-sweep candidates (group-conformal FDR$\leq$0.05) could be genuinely undiscovered strong lenses, via an expanded crossmatch (SIMBAD + DES/KiDS VizieR; NED/SuGOHI unreachable) and two methodologically-independent visual passes: a structured Huang-VI rubric pass with an adversarial skeptic re-check, and the independent `lensjudge` harness (factored multiagent, Claude-opus). *Qualified* = still NEW after the crossmatch AND graded A/B by BOTH passes.

Result. Of 737 candidates, 136 (18%) coincide with a known lens or lens-candidate (92 DES, plus KiDS and SIMBAD lens-types); **601** remain NEW. *None* of the 601 new candidates is graded A/B by either grader's strongest pass, so **qualified** = 0. This is not the graders rejecting everything: together they graded 5 candidates $\geq$B, and **all 5 are already-catalogued lenses** (4 sub-arcsecond DES matches) — the consensus re-discovered real lenses and found nothing comparable among the new ones.

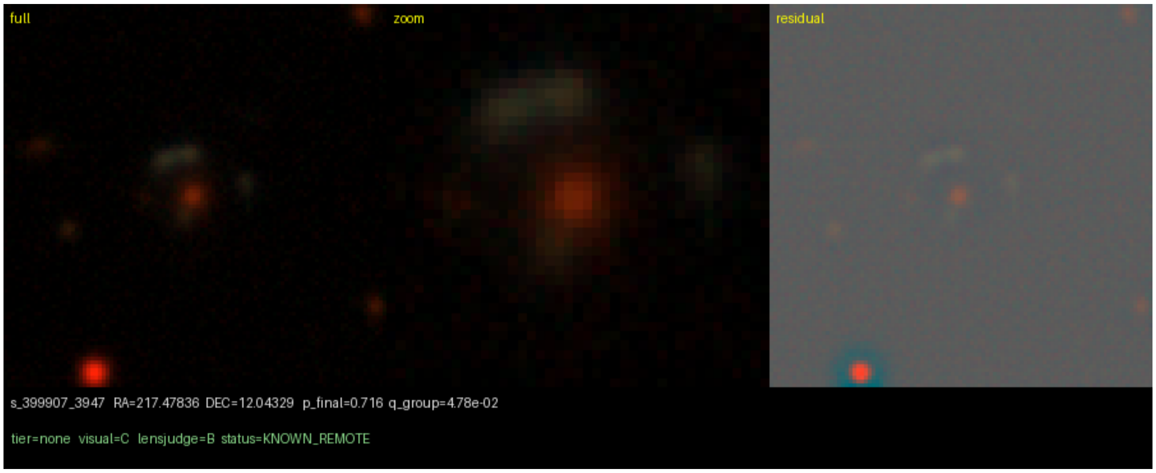

Figure 3: s_399907_3947: RA=217.47836, DEC=12.04329, p_{final}=0.716; status KNOWN_REMOTE (nearest curated at 6464.70″); visual C, lensjudge B. Full | zoom | residual.

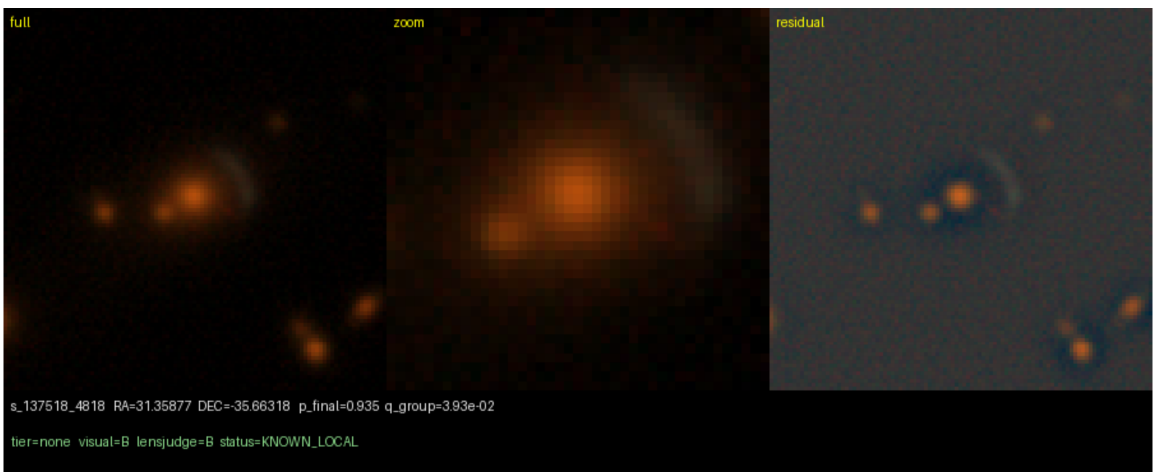

Figure 4: s_137518_4818: RA=31.35877, DEC=-35.66318, p_{final}=0.935; status KNOWN_LOCAL (nearest des_jacobs2019 at 0.14″); visual B, lensjudge B. Full | zoom | residual.

Lenses the campaign re-discovered (already catalogued; the validation):

7 ClaudeNet v3: contaminant-aware lens finding and Euclid cross-validation

The v2 DR9 sweep ended in a decisive negative (§6): of 601 genuinely-new group-conformal candidates, *zero* were graded A/B by either independent vetter, and the only five graded ≥B were already-catalogued lenses. The candidates were overwhelmingly luminous-red galaxies with companions, rings, and blends. This diagnoses the binding constraint precisely: it is not architecture (the v1/v2 finding) but *lens-vs-mimic separation* and *vetting resolution*. The field-wide remedy is the same — contaminant-targeted hard negatives give an 11× false-positive reduction at 2.3% true-positive cost (MNRAS, 2025). v3 builds a contaminant-aware finder, deploys it on DR10/DR11, and cross-validates the overlap against Euclid Q1. We keep all v1/v2 numbers; v3 is additive, and every number below is read from a tracked artifact under `data/v3/` or `v3/`.

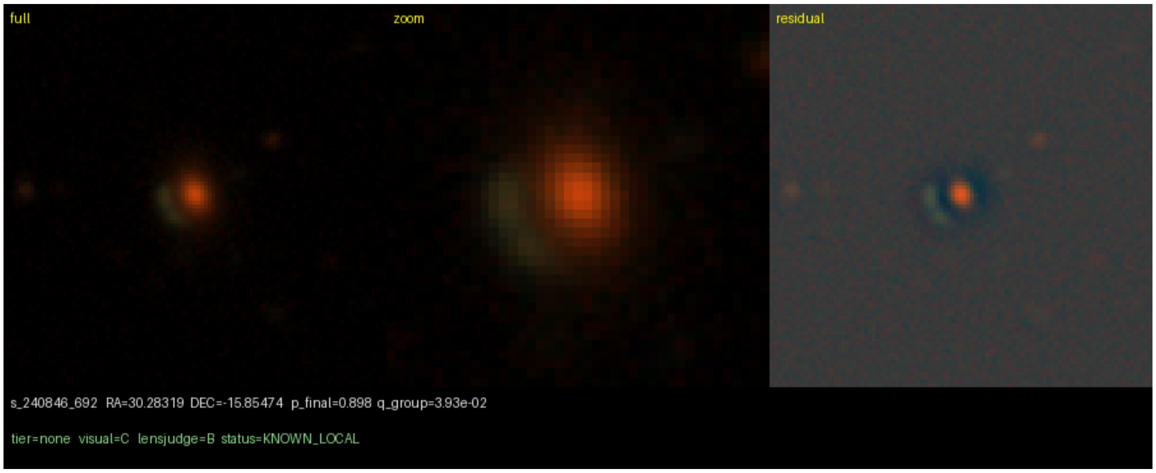

Figure 5: s_240846_692: RA=30.28319, DEC=-15.85474, p_{final}=0.898; status KNOWN_LOCAL (nearest des_jacobs2019 at 0.05″); visual C, lensjudge B. Full | zoom | residual.

Figure 6: s_137502_4157: RA=26.44498, DEC=-35.69096, p_{final}=0.892; status KNOWN_LOCAL (nearest des_jacobs2019 at 0.19″); visual C, lensjudge B. Full | zoom | residual.

The mimic metric and the v2 baseline (A0). The v1/v2 headline measured recovery at a false-positive rate set against *random* galaxies; the campaign showed that is the wrong null. We define **recovery@matched-mimic-FPR**: identical arithmetic, but the negative class is a bank of lens-*mimics* rather than random galaxies. Against this null the shipped v2-lean ensemble — which recovers 96–100% of held-out lenses at random-FPR — recovers only 0.168 (Storfer) / 0.307 (Inchausti) at mimic-FPR 0.05 on the curated 601-mimic seed bank. The mimic-discrimination gap, not random-galaxy recovery or architecture, is what v3 must close.

The lens-mimic bank (A1). Rather than mine a fresh pool we reuse the DR10 sweep: 148,034 NEW survivors are CNN-high, DR10-native non-lenses, morphology-typed by joining the Tractor catalogue (extended-LRG 34 k — the MNRAS (2025) regime — plus compact-REX, exp-disk, and the campaign's fine types). A 120-row stratified agentic G1 gate confirms 98.3% non-lens purity. The bank is 118,901 training rows + a frozen, stratified 29,724-row held-out mimic-eval that never enters training.

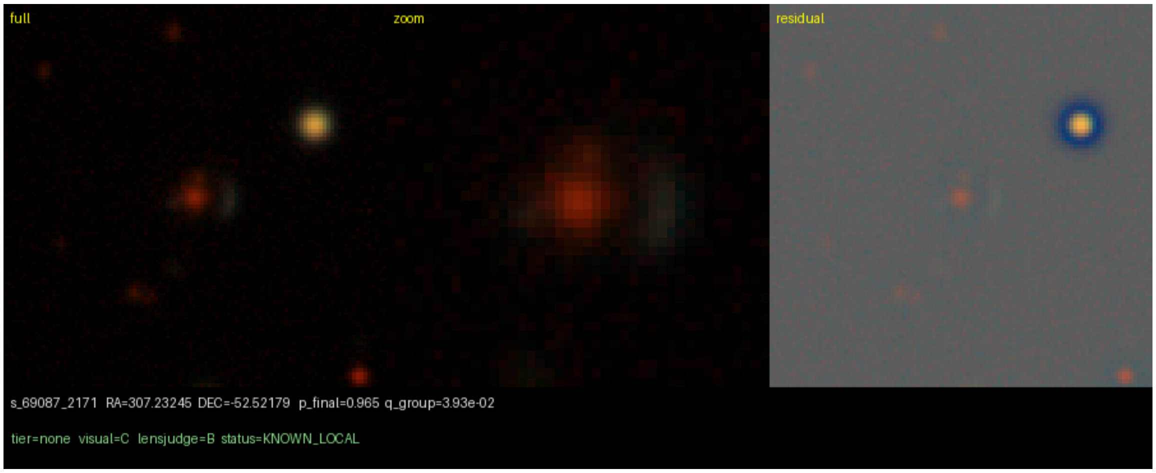

Figure 7: s_69087_2171: RA=307.23245, DEC=-52.52179, p_{final}=0.965; status KNOWN_LOCAL (nearest des_jacobs2019 at 0.09″); visual C, lensjudge B. Full | zoom | residual.

Table 6: A2 contaminant-aware retraining (mimic-blend f=0.5), v2-lean→v3blend8, on the held-out DR10 mimic-eval. Mimic-recovery rises at every operating point; the random-galaxy recall (testneg null) drops only slightly (the staf327 trade).

	recovery@mimic-FPR (Storfer)		recovery@random-FPR(0.01)	
	@0.05	@0.01	Storfer	Inchausti
v2-lean	0.713	0.460	0.967	0.998
v3blend8	**0.793**	**0.699**	0.935	0.979

Contaminant-aware retraining (A2). We retrain the three swappable members with the v2 "hard" recipe held *byte-identical* (same n_{mine}=10,000 displaced bootstrap negatives, same per-member seed) and change only the negative *content*: a fraction f of the displaced negatives become typed mimics from the bank. A blend-fraction sweep selects f=0.5 (f=0.7 over-specialises). On the held-out DR10 mimic-eval (Table 6) v3blend8 lifts Storfer mimic-recovery from 0.713→0.790 at 0.05 and 0.460→0.699 at 0.01, with the largest per-type gains on the dominant lrg_companion/extended-LRG contaminants. The cost is a small, real drop in *random*-galaxy recall (0.967→0.935 Storfer at random-FPR 0.01) — exactly the MNRAS (2025) trade.

Ensemble refit and the lens-vs-mimic head (A6, A3). A roster search over the average combiner finds **v3blend8** — keeping *both* the v2-hard and v3-mimic-blended version of each member — is near-Pareto: it recovers random-FPR recall to within CI of v2-lean while roughly doubling mimic-recovery. (The 194 K-free l18 ResNet is, by contrast, dead weight for mimic separation; dropping it maximises mimic-recovery at a 3% random-recall cost.) A learned logistic *lens-vs-mimic head* on the eight member scores tops every average roster at the ϕ=0.05 operating point, but it overfits the strict tail and — as §7 shows — does not generalise to the independent Euclid test, so the deployable model is **v3blend8**, with the head as a ϕ=0.05 stage-2 re-ranker.

DR10/DR11 deployment and the discovery result (C, D5). We swept DR10-south (43.7 M parent galaxies) with v2-lean as the head-to-head baseline and re-ranked the 150 k survivors with v3blend8 + the head. *Recall is preserved*: the known Inchausti-811 lenses sit at the 96$^{\mathrm{th}}$ percentile

Table 7: C-vet agentic grading of the top-30 NEW candidates (same harness, `rubric_imaging_v2`). v3's mimic-aware ranking does not raise the top-of-list A/B yield over v2 at DECaLS resolution (the discovery frontier is resolution-bounded); the D's stay lrg_companion-dominated.

ranking	A	B	C	D	A/B
v2 p_final (DR10)	2	6	11	11	**8**
v3-head (DR10)	2	3	11	14	**5**
v3blend8 (DR10)	2	5	9	11	**7**

Table 8: DR10 vs DR11 on the same EDF-F/S DECam sky: v3blend8 region-local recovery of Euclid grade-A lenses into the top-K, and the cross-validated NEW-confirmed counts. DR11's deeper reductions give a small, consistent edge; v3 transfers cleanly across DECam releases (contrast the DR11-north BASS/MzLS failure, D3).

release	Euclid-A in top-100	in top-1000	NEW-confirmed A/B
DR10 (D4)	4	7	41
DR11 (D5)	5	10	41

Cross-release (by Euclid id): 33 shared, 49 unique cross-validated lenses.

of the survivor pool by the v3 score, and ∼88% of their grade-A are found at high score (the operating-point recall gap is a DR9→DR10 domain shift, not a model failure). As at DR9, full-m conformal selection certifies *nothing* at this sweep scale — a surfaced power-floor limit, not a null result. The candidate list therefore comes from top-by-score ranking, vetted agentically. Table 7 is the honest discovery result: v3's mimic-aware ranking does *not* raise the top-30 A/B yield over v2 (7 vs 8), and the rejects stay lrg_companion-dominated — because the top of the NEW-survivor pool is intrinsically mimic-dominated at DECaLS 0.262″/px, where $\theta_E \sim 1\text{–}\ 2''$ is 4–8 px. A DR11-south re-run on the same DECam sky behaves identically, with a small consistent edge from the deeper reductions (Table 8).

Euclid Q1 cross-validation: the resolution lever (D1, D4, D5). Where DESI overlaps Euclid Q1 (Euclid Collaboration, 2025) (0.1″/px, ∼13× DESI; the three Deep Fields), we can both validate v3 against an *independent* expert-graded lens sample and break the resolution wall. Two results. *(i) v3 is grade-selective for real lenses:* it recovers Euclid grade-A lenses into its survivor pool at 7.1% versus grade-C at 0.8% — an 8.6× enrichment (Fig. 8, right) — confirming on an independent sample that it finds genuine lenses. But absolute recall is capped: ∼40% of Euclid grade-A/B lenses are not even in the DESI parent (too faint/small for DESI; Fig. 8, left), a depth/resolution ceiling no model can cross. *(ii) Escalation converts the overlap:* grading the v3∩Euclid cross-matches with the LensJudge v2 escalation grader (§7) at 0.1″ flips them from DESI-ambiguous to confirmed — median p_{lens} 0.03 → 0.70 on 134 real objects (Fig. 8), with our grades agreeing with the Euclid experts on ∼90% of grade-A. The yield is **49 cross-validated A/B candidates** across DR10+DR11 (Table 11, Appendix A; 33 recovered in both releases, 8 DR11-unique). We are explicit: these are NEW to the DESI lens-finder catalogues but *are* in the published Euclid Q1 catalogue — v3 independently *recovers* real Euclid lenses from DESI imaging, a cross-validation of both pipelines, *not* net-new discoveries.

Honest limits and a selection-bias correction (D6, D3). The v3-vs-v2 mimic-separation gains above are real, but the cross-model comparison demands one correction (Table 9). The seed

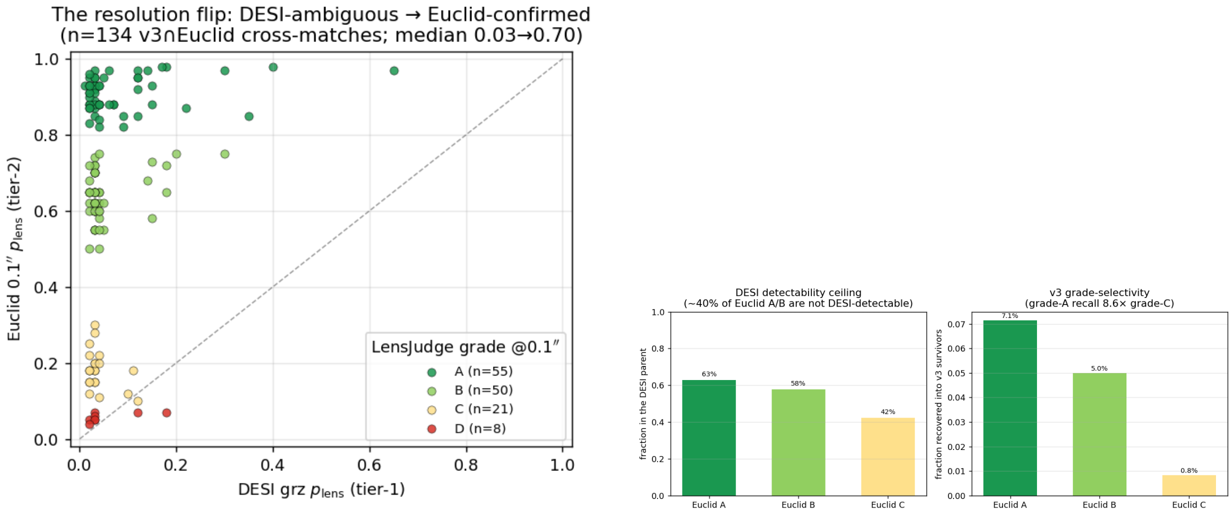

Figure 8: The resolution lever. Left: the DESI→Euclid p_{lens} flip on the real v3∩Euclid cross-matches — nearly every object is DESI-ambiguous ($x \lesssim 0.2$) but Euclid-confirmed ($y \gtrsim 0.6$). Right: Euclid grade-resolved recall — the DESI detectability ceiling (~40% of Euclid A/B missing from the parent) and v3's 8.6× grade-A-over-C survivor selectivity.

Table 9: Candidate quality across model generations (D6). recovery@matched-mimic-FPR(0.05) and the *independent* Euclid grade-A-vs-C AUC. The seed column is selection-biased (the seed bank is what the v2/v3 ensemble scored high), so orig-vs-v3 there is invalid; the trustworthy comparison is the Euclid AUC, where all models sit within ~1 SE (≈0.06, n_A=66).

model	seed mimic@0.05 (Storfer)	DR10-eval@0.05 (Storfer)	DR10-eval@0.05 (Inchausti)	Euclid A-vs-C AUC (indep.)
orig (random-neg CNN)	0.592	0.598	0.640	0.613
v2-lean	0.168	0.713	0.856	0.647
v3blend8	0.312	0.790	0.896	0.660
v3-head	0.647	0.885	0.922	0.615

mimic-FPR metric is *selection-biased*: the seed bank *is* the set the v2/v3 ensemble scored high, so it saturates their threshold and deflates their recovery, while a model less central to the selection (a single random-negative EfficientNet) looks artificially strong there — so any "$N\times$ over the original published models" read off the seed is invalid and we do not claim it. The trustworthy comparison is the *independent* Euclid grade-A-vs-C AUC: the original CNN, v2-lean, v3blend8, and the head all sit at 0.61–0.66, within ~1 standard error (n_A=66) — statistically indistinguishable at the hardest distinction. v3 *does* clearly beat the original on the broad mimic population (DR10-eval recovery 0.60 → 0.89), i.e. it yields a meaningfully cleaner candidate list to vet, but it has no significant edge at the resolution-bounded discovery frontier. Finally, v3 is **DECam-specific**: a DR11-*north* (BASS/MzLS) sweep shows systematic under-scoring and zero Euclid-A/B in its top-30 — north-footprint deployment needs instrument- native retraining.

Vetting: LensJudge v2 (summary). The Euclid cross-validation above rests on the v2 upgrade of the `lensjudge` agentic vetter, summarised here (full detail in `lensjudge/V2_FINDINGS.md`). Three changes carry it: (i) a *two-tier high-res escalation* (DESI grz tier-1, Euclid 0.1″ tier-2 when coverage exists), live-validated as the p_{lens} flip; (ii) a rubric tuned to the dominant failure modes

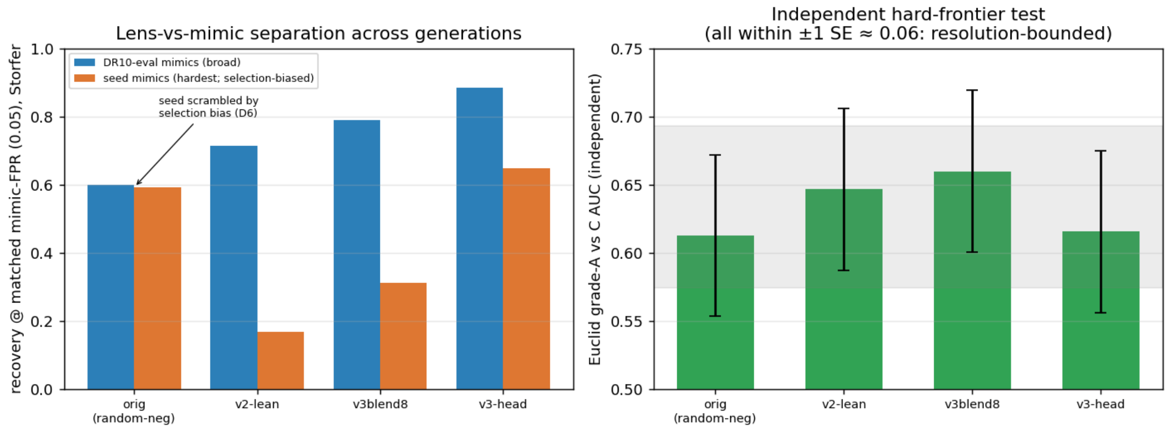

Figure 9: Candidate quality across model generations. Left: recovery@matched-mimic-FPR(0.05) on the broad DR10-eval mimics (clear orig→v2→v3 progression) and on the curated seed mimics (scrambled by the selection bias — the seed is what the v2/v3 ensemble scored high). Right: the *independent* Euclid grade-A-vs-C AUC — all four models lie within ∼1 standard error, i.e. the hardest real-vs-mimic distinction is resolution-bounded for every model.

(LRG+companion/ring/spiral/blend), which lifts lens-vs-mimic discrimination by +0.196 AUC over the v1 grader on a frozen labelled benchmark; and (iii) an agency ablation showing the agentic loop is unnecessary for *detection* (a one-shot direct grader matches it 6× cheaper) but essential for *mimic adjudication* — motivating the deployed cascade (cheap direct triage + agentic v2 on survivors + Euclid escalation by coverage).

8 ClaudeNet v4: mean-combiner selection and release-native fine-tuning

Deploying `v3blend8` on the final release (DR11-south, §7) surfaced an apparent collapse: known-lens grade-A recall at the per-member 10^{-4} operating point fell 62% → 41% relative to DR10. v4 diagnoses this as an *operating-point* artifact — not a model failure — and fixes it with two stacking, release-portable levers: a combiner change (free) and a release-native fine-tune (one short GPU pass). v4 is additive — it keeps the v3 contaminant-aware members and adds (i) a selection rule and (ii) DR11-native weights.

The recall collapse is the combiner, not the model. The stage-1 selector was a per-member 10^{-4} *union*; on the deeper DR11 imaging the per-member thresholds tighten and the `resnet46_C` member saturates, so the union is dominated by single-member spikes. Three facts show the model is intact: (i) 21/90 of the grade-A lenses lie outside the searched footprint or are cut by the parent selection, so the *searchable* recall is already 54%, not 41%; (ii) the threshold-free AUC of in-parent held-out grade-A versus a 2M random DR11 negative sample is **0.9955** for the five-member calibrated mean — the DR9-trained members still rank DR11 lenses far above random DR11 galaxies; (iii) replacing the union with the calibrated five-member *mean* at the *same* survivor budget recovers grade-A recall 54%→75% (88% at 150k) and reaches DR10's in-domain 62% at fewer than 43k survivors. The calibrated mean is therefore the default stage-1 selector going forward (`162/370 --stage1-combiner mean`); it carries no per-release threshold recalibration, removing the DR9→DR11 drift entirely.

Table 10: v4 fine-tune gate: held-out (training-excluded) Storfer/Inchausti recall at the matched-FPR 95k-survivor operating point, calibrated-mean combiner. Baseline is the v3 five-lean ensemble; v4 swaps the three members for their DR11-native fine-tuned versions. The release-native fine-tune cracks the hard residual the resolution-limited v3 model could not.

held-out set	v3 baseline	v4 fine-tuned	Δ
Inchausti grade-A	0.724	**0.816**	+0.09
Inchausti grade-B	0.515	**0.812**	+0.30
Storfer grade-A (hard)	0.544	**0.796**	+0.25

A release-native training pool. The v1–v3 positive set was capped at 1,961 DR9-pixel lenses. We broadly re-harvested the confirmed-lens literature for the DR11-south footprint — reclaimed VizieR/DES/KiDS/HSC catalogs, SLACS/BELLS spectroscopic lenses, lensed-quasar compilations, a bulk SIMBAD object-type pull, and the Euclid-Q1 grade-A/B set — deduplicated at 5″ and tiered by confidence, yielding 3,171 **net-new high-confidence (gold+silver) positives**, a $\sim$13$\times$ expansion of the DR11-south pool. Storfer and Inchausti are held out for the recall metric throughout (excluded as both positives *and* negatives), so the gate below measures genuine generalisation, not memorisation.

Release-native warm-start fine-tune. All positive cutouts are re-extracted DR11-south-native (`griz`, 101 px), tier-subsampled to 5,806, and paired with 30k random + 20k *hard* DR11 negatives (CNN-high mimics drawn from the mean-ranked survivor pool, PU-guarded and candidate-region-skimmed). The three swappable members (`effnet_S2`, `effnet_B3`, `resnet46_C`) are warm-started from their `_b50` checkpoints and fine-tuned for 12 epochs at lr=3×10^{-4}; `effnet_B` and `zoobot_N` stay frozen as the random-FPR / diversity anchors. The gate on held-out Storfer/Inchausti at DR11 resolution (mean combiner, matched-FPR) passes decisively, with the largest gain on the lrg+companion hard residual that the combiner alone cannot touch (Table 10); AUC does not regress (Inchausti-A 0.996$\rightarrow$0.998).

Full re-sweep and the v4 candidate set. We re-scored all 5.38×10^7 DR11-south galaxies with the fine-tuned ensemble — reusing the original parent cutouts (no re-extraction) and combining the mean of [`effnet_B`, `zoobot_N`, the three `_b50_dr11` members] — and selected the top 150k by mean. Held-out recall (position-crossmatch; Storfer/Inchausti training-excluded) is the best of any configuration: Inchausti grade-A **0.87** (60/69), grade-B **0.87** (76/87), Storfer grade-A **0.825** (85/103), versus 0.54/0.32 for the original union-95k operating point. The full re-sweep recovers the out-of-pool lenses a survivor-only re-rank cannot: the v4 top-150k retains only 15,922 of the union-95k survivors and **adds** 134,078 — a substantially new, recall-richer candidate pool.

Honest scope. v4 *stacks* domain adaptation on v3's mimic rejection; both levers are release-portable — the recipe (re-harvest the native-resolution positive pool, mine native hard negatives, warm-start fine-tune, select by mean) is precisely what transfers to the higher-resolution surveys that are the program's endpoint. v4 remains DECam-specific (the northern footprint needs BASS/MzLS-native retraining, §9), the DECaLS 0.262″/px resolution ceiling is unchanged (higher-resolution vetting remains the decisive net-new-discovery lever), and the v4 candidate set is recall-richer but *pending* vetting — the completed as-run DR11-south vetting used `v3blend8`.

9 Future directions

The campaign localises two distinct limits with concrete, separable levers. The discovery frontier is bounded by *vetting resolution* (model improvements saturate at DECaLS 0.262″/px), and the footprint is bounded by *training instrument* (v3 is DECam-specific). The priorities follow directly.

1. Euclid pre-screening (the decisive discovery lever). The cross-validation (§7) shows the productive mode: v3 pre-screens the DESI imaging, Euclid confirms at 0.1″. Q1's $\sim$63 deg^2 over three Deep Fields overlaps only $\sim$0.5% of a DESI-south sweep, which is why net-new yield there is bounded; *Euclid DR1* (wide-area) inverts this. The ready action now is a targeted sweep restricted to the Euclid-overlap sky, where *every* candidate is escalatable — the one configuration in which v3's separation and the resolution- breaking vet both apply to every object.

2. BASS/MzLS-north retraining. v3 does not transfer to the northern third of the sky (D3). Mining a BASS/MzLS-native lens-mimic bank and retraining the members (the v3 recipe is instrument-agnostic; only the negatives change) would unlock DR9/DR11-north and roughly triple the addressable footprint.

3. *i*-band / foundation-model revival for small-θ_E. DR10/DR11-south are native *griz*. The deferred AION-1/i-band lever (A4) targets the red-lens/red-companion degeneracy and the small-θ_E bin specifically; it is worth re-testing under the narrower lens-vs-mimic task (rather than the random-FPR task where it failed in v2) on the native-i rows the bank already carries.

4. A pixel-level lens-vs-mimic head. The learned logistic head on member scores beats the average at ϕ=0.05 but overfits the strict tail and does not generalise to the independent Euclid ranking. A small CNN head trained directly on pixels with positives=lenses, negatives=the typed mimic bank (rather than on the eight member scores) is the natural next step, used as a post-conformal re-ranker.

5. The active-learning loop. The `lensjudge` harness can export confident-D rejections (true mimics, *not* objects that flipped to A/B at higher resolution) as fresh, typed hard negatives for the next-generation bank — closing a mine→grade→retrain loop that continually sharpens the dominant contaminant classes.

6. A full DR11-south candidate catalogue. *Realised — see §8.* v3blend8 runs on DR11-south with essentially no adaptation (the pipeline is release-parametrised); v4 went further — replacing the per-release threshold recalibration with a calibrated-mean selector and adding a release-native warm-start fine-tune — and re-swept the full footprint into the v4 candidate set. The honest caveat stands: net-new yield is resolution-bounded, so the v4 catalogue is recall-richer but its new candidates remain pending higher-resolution vetting.

Deployment recommendation. For DR11-south, ship the **v4** release-native fine-tuned ensemble with the **calibrated-mean stage-1 selector** (§8); retain **v3blend8** for DR10-south and the Euclid cross-validation. In all cases use the score *mean* (not the learned head, which does not generalise past ϕ=0.05) and the LensJudge cascade (direct triage + agentic mimic adjudication + HSC/Euclid escalation by coverage) for vetting. This keeps the v2-lean lens recall, adds v3's contaminant-rejection edge and v4's release-native gain on the hard residual, and routes the resolution-breaking confirmation to exactly the candidates with higher-resolution coverage.

10 Conclusion

Across three generations the lesson sharpened. v1 showed the lever the lineage left unused is member *diversity*, not the backbone, and won cleanly at every matched-FPR operating point. v2 confirmed this at deployment scale and produced the first FDR-controlled sky sweep — whose qualification campaign then delivered the most informative result of the program: the binding constraint is *lens-vs-mimic separation* and *vetting resolution*, not architecture. v3 answers the first half with a contaminant-aware finder that rejects common mimics far better than any prior ClaudeNet model; the Euclid cross-validation answers the second — at DECaLS resolution every model, old and new, is statistically tied at the hardest real-vs-mimic distinction, and the only operation that converts candidates to confirmations is higher-resolution imaging. The honest, repository-consistent takeaway is that for DECaLS lens finding the returns are in data, calibration, and contaminant-aware negatives — a meaningful but not transformational model lift — while the decisive levers for net-new discovery are *resolution* (Euclid pre-screening) and *footprint* (instrument-native retraining), the directions of §9. **v4** (§8) takes the data/calibration lever as far as it goes on DR11: a calibrated-mean stage-1 selector erases the apparent DR9→DR11 recall collapse for free, and a release-native warm-start fine-tune — for the first time — cracks the lrg+companion hard residual (held-out grade-A recall to 0.87, hard-residual 0.54→0.83), establishing the release-native recipe the higher-resolution surveys will reuse.

A Cross-validated candidate catalogue

Table 11 lists the 49 unique DESI v3 survivors that match a Euclid Q1 discovery-engine lens, are NEW to the DESI lens-finder catalogues (Storfer/Inchausti/Huang within 5″), and were graded at Euclid 0.1″ by the LensJudge v2 escalation grader. These are independently *recovered* Euclid catalogue lenses — a four-way cross-validation (DESI v3 CNN ∩ Euclid discovery engine ∩ Euclid expert grade ∩ LensJudge-at-0.1″) — and are deliberately *not* claimed as net-new discoveries. The p_{lens} column shows the DESI→Euclid tier flip; "release" marks whether the object was recovered in the DR10 sweep, the DR11-south sweep, or both.

Table 11: Cross-validated strong-lens candidates: DESI v3 survivors that match a Euclid Q1 discovery-engine lens, NEW to the DESI lens-finder catalogues (Storfer/Inchausti/Huang), graded at Euclid 0.1″. These are independently recovered Euclid catalogue lenses, *not* net-new discoveries. p_{lens} shows the DESI→Euclid tier flip.

DESI id	RA (deg)	Dec (deg)	Euclid grade	LensJudge @0.1″	p_{lens} DESI→Euclid	release
s_83157_8409	57.7025	-48.4062	A	A	0.12→0.97	both
s_85094_13993	63.6294	-48.0333	A	A	0.30→0.97	DR10
s_79373_7769	58.2019	-49.4705	A	A	0.05→0.95	both
s_80329_6402	64.5419	-49.3086	A	A	0.03→0.95	both
s_81272_7726	63.9239	-48.9296	A	A	0.03→0.93	both
s_86048_6517	58.6493	-47.6453	A	A	0.02→0.93	DR10

DESI id	RA	Dec	Euclid	LensJudge	p_{lens}	release
s_169188_5518	53.3251	-29.1510	A	A	0.15→0.93	both
s_179335_8072	51.0363	-27.2415	A	A	0.01→0.93	both
s_84136_4630	65.1956	-48.3600	A	A	0.07→0.88	DR10
s_174235_13613	51.5787	-28.1897	A	A	0.02→0.88	both
s_178058_11508	51.9633	-27.4260	A	A	0.04→0.88	both
s_165427_7416	52.9210	-29.9100	A	A	0.03→0.87	both
s_175513_5175	53.5621	-28.0129	A	A	0.02→0.87	DR11
s_82216_7491	62.1645	-48.8224	A	A	0.09→0.85	both
s_93968_433	62.8684	-45.7047	A	A	0.04→0.84	both
s_170450_15146	54.4805	-28.9549	A	A	0.04→0.82	both
s_81263_7090	60.4836	-49.0669	A	B	0.15→0.73	both
s_171714_3810	54.7757	-28.6645	A	B	0.02→0.72	DR11
s_90958_9241	59.7052	-46.5695	A	B	0.03→0.70	both
s_176778_4403	50.8798	-27.6486	A	B	0.03→0.70	both
s_170452_14504	54.9726	-29.0757	A	B	0.18→0.65	both
s_84129_16461	62.7892	-48.2719	A	B	0.04→0.62	both
s_88015_3655	65.1247	-47.3087	A	B	0.05→0.62	both
s_88988_8317	62.3367	-47.0669	A	B	0.03→0.62	both
s_167928_8112	52.3902	-29.5927	B	A	0.02→0.95	both
s_84129_17659	62.8043	-48.2286	B	A	0.03→0.93	DR10
s_85093_15882	63.2503	-48.0737	B	A	0.04→0.93	both
s_77522_7961	63.7748	-50.0119	B	A	0.02→0.91	both
s_180619_719	51.2423	-27.0221	B	A	0.02→0.83	both
s_179345_702	53.6441	-27.2233	B	B	0.03→0.74	both
s_86061_15501	63.6862	-47.8550	B	B	0.03→0.72	both
s_166673_9358	51.7071	-29.7125	B	B	0.03→0.72	both
s_77510_6177	59.0844	-49.9562	B	B	0.02→0.65	DR10
s_80314_12621	59.0815	-49.3650	B	B	0.04→0.65	both
s_78453_11482	63.9706	-49.7278	B	B	0.02→0.65	both
s_84123_2953	60.2940	-48.3238	B	B	0.03→0.65	both
s_88997_602	65.3650	-47.0251	B	B	0.04→0.65	both
s_81263_7679	60.4929	-48.9067	B	B	0.03→0.62	DR10
s_87026_5695	60.5672	-47.6155	B	B	0.02→0.62	DR10
s_88007_7987	62.2189	-47.2949	B	B	0.03→0.62	DR10
s_84125_13400	61.3553	-48.2723	B	B	0.04→0.60	DR11
s_88996_2557	65.0706	-47.0601	B	B	0.02→0.60	DR11
s_167935_8040	54.3727	-29.3848	B	B	0.04→0.60	DR11
s_75664_6939	58.9597	-50.5340	B	B	0.04→0.55	DR11
s_81254_8721	57.2304	-49.0460	B	B	0.03→0.55	DR11
s_88990_5749	63.0160	-47.0193	B	B	0.03→0.55	both
s_176778_10456	51.0181	-27.7804	B	B	0.03→0.55	both
s_76594_8452	62.6149	-50.3258	B	B	0.04→0.50	DR11
s_176791_230	54.4517	-27.8613	C	A	0.03→0.93	both

B Lineage-NEW grade-A survivors are literature rediscoveries

Table 12 lists the v3 grade-A candidates surfaced in the DR10 vetting that the lineage crossmatch (`_clib.known_lens_catalogs`: Storfer 2024, Inchausti 2025, Huang 2020/2021) found absent from the catalogued sample. We then ran a literature-wide NED/SIMBAD crossmatch (`94_external_xmatch.py`) — the step the lineage-only check had skipped — and it is decisive: *every*

Table 12: v3 grade-A C-vet survivors that are *absent* from the Huang/Storfer/Inchausti lineage catalogues, with their literature-wide NED/SIMBAD crossmatch (`94_external_xmatch.py`). All three are previously-published lenses from other surveys — v3 *rediscoveries*, not net-new. (A fourth C-vet survivor, `s_310364_6649`, is the Huang+2020 lens `DESI-038.2078-03.3906`.) Net-new lenses: zero; the result is independent external validation of the v3 finder.

DESI id	C-vet $p_{\rm lens}$	ranking	known lens (survey)	sep
s_441355_1111	0.74	head	CSWA 1 (CASSOWARY)	$0.5''$
s_124958_10481	0.82	v3blend8	DES J035242.40-382544.9 (DES)	$0.2''$
s_102952_9916	0.76	v3blend8	DES J221912.39-434835.1 (DES)	$0.2''$

survivor is a previously-published strong lens. `s_310364_6649` is the Huang+2020 grade-A lens `DESI-038.2078-03.3906` ($0.2''$); the other three coincide ($\leq 0.5''$) with **CSWA 1** (CASSOWARY; Belokurov et al. 2009), **DES J0352−3825** (Dark Energy Survey; O'Donnell et al. 2022), and **AGEL J221912−434835** (AGEL; Tran et al. 2022, also in DES). The honest count of *net-new* lenses is therefore **zero**. This is not a null result: v3 *independently recovered* four genuine, separately-confirmed lenses across four surveys at DESI resolution — a real external validation of the finder — and it is why the known-lens set now folds in Huang+2020 and a literature catalogue (`external_lens_catalog.csv`) so a published lens can no longer be miscounted NEW. The positions still have *no* Euclid coverage; net-new discovery at DESI resolution remains resolution-bounded (§7).

C Reproducibility

All results are reproduced from tracked code and artifacts on branch `reproductions/claudenet-v3`. The script numbering is: `00--91` v1 (members, calibration, combiners, the seven-phase program, tables/figures); `100--165` v2 (NegEval-1M, mining, refit, the DR9 sweep, group-conformal selection); `200--260` the DR9 qualification campaign; and `300--367` v3 — `300--326` workstream A (mimic metric, bank, retraining, ensemble refit, lens-vs-mimic head), `360--367` deployment (DR10/DR11 parents, scoring, selection, the Euclid cross-validation and the model-progression comparison). The vetting harness lives under `reproductions/lensjudge`. Shared libraries (`_ensemble`, `_train`, `_clib`, …) reuse the baseline matched-FPR arithmetic verbatim, so every v1/v2/v3 number is directly comparable. The v3 figures and tables in this report are regenerated by `92_make_v3_figures.py` and `92_make_v3_tables.py` from the artifacts under `data/v3/` and `v3/`.

Survey-scale compute ran on NERSC Perlmutter (GPU account `cosmo_g`, CPU `cosmo`); the v3 program used $\mathcal{O}(10^2)$ GPU-hours. Agentic vetting (LensJudge) spent ∼\$53 of a \$100 cap, targeted on the most-promising candidates with an evidence gate. We carry the lineage's honesty discipline throughout: "candidate" never "discovery"; recall of the published catalogues is reported before any new-candidate count; both independent graders are Claude models (different harness/prompt, not statistically independent); the mimic-FPR seed metric is selection-biased and its cross-model reading is corrected against the independent Euclid measure (§7); and DR11 results are reported embargo-aware (internal until the DR11 public release).

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